BA305 Project Proposal — Fall 2026

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0. TEAM REPOSITORY

- **https://github.com/jberisso/BA305-A1---hotel-bookings**

1. DATASET

- **Dataset #15 - Hotel booking records**

2. PREDICTIVE QUESTION

- **How well does lead time predict booking cancellations, and does its predictive value vary across combinations of country and market segment?**
- **We intend to estimate the is_cancelled variable.**
- **It is a classification.**
- **Hotel and resort managers could use these predictions to anticipate cancellations and decide how much booking availability to keep. This could inform overbooking decisions, helping managers balance the risk of empty rooms against the risk of accepting more guests than they can fit.**

3. DESCRIPTIVE QUESTION

- **Do bookings at the city hotel and resort hotel have distinct profiles based on lead time, price, party size, and special requests, or are they more similar when these characteristics are considered together?**
- **Clear separation would suggest that the two properties attract different booking profiles. Overlap between the two would suggest that their bookings share more characteristics than their different property types might imply.**
- **Managers overseeing both properties could use these findings to assess whether shared pricing and marketing strategies are appropriate or whether each property needs a different approach.**

4. WHAT YOU ALREADY SEE IN THE DATA

- **For 81 of the 87 agents examined, every nonrefundable booking was canceled. This pattern warrants investigation, although it does not establish that the records are incorrect.**
- **Country, agent, and company information is missing from approximately 0.4%, 14%, and 94% of booking records, respectively.**
- **Approximately 63% of bookings were not canceled, and 37% were canceled, indicating moderate class imbalance.**

5. RISKS

- **Unjustified data removal could bias our findings. We need a defensible basis for deciding whether the unusual cancellation patterns reflect data quality issues or actual booking behavior. Removing records solely because they contradict our expectations could distort the analysis.**
- **The findings may not generalize to other hotels. The data covers two Portuguese hotels from 2015 to 2017, so the patterns may not apply to other locations or periods.**
- **Lead time may have similar predictive value across countries and market segments. This would still answer our question, but it would limit the case for tailoring predictions by country and segment.**

6. WHAT YOU WANT FROM US

- **We noticed that a handful of booking agents have every single one of their non-refundable bookings end up canceled. We want to ask how to be sure that we're allowed to say 'these look suspicious' and remove them from our data, since we've only learned charts and tables so far, and not a formal way to define suspicious data?**